ROMAN NALAPKO

JUNIOR PYTHON BACKEND & AUTOMATION ENGINEER | APIs · PostgreSQL · AI Integrations

Izmail, Ukraine | Remote within Ukraine | itromanalapko@gmail.com | LinkedIn | GitHub

PROFILE

Junior Python backend and automation engineer with hands-on delivery of a private client workflow and an open-source multi-tenant RAG platform. Builds asynchronous APIs, third-party and LLM integrations, validated data flows, Dockerized services, observability, automated tests, and CI.

TECHNICAL SKILLS

Backend Python, FastAPI, asyncio, REST APIs, SSE, webhooks, Pydantic, SQLAlchemy, Alembic

Automation and AI API integrations, workflow automation, structured model output, Gemini API, Whisper, RAG, hybrid retrieval, embeddings

Data and Platforms PostgreSQL, SQL, Qdrant, Cloudflare Workers, D1, KV, Docker, Docker Compose, Linux

Quality and Operations pytest, GitHub Actions, CI/CD, Prometheus, structured logging, health checks, failure handling, Git

SELECTED ENGINEERING WORK

Private Client - AI Finance Assistant | *Python, PostgreSQL, Gemini, Whisper*

- Delivered a private finance workflow with a Telegram interface for text and transcribed voice input, covering data modeling, integrations, validation, testing, Docker configuration, and technical handoff.
- Converted probabilistic model responses into validated application data and required explicit confirmation before persisting amounts, currencies, and categories.
- Implemented user-scoped queries, Alembic migrations, atomic writes and rollback behavior, currency-separated summaries, XLSX export, service health checks, and automated parsing and persistence tests.

AI RAG Platform | *FastAPI, PostgreSQL, Qdrant, Docker* | GitHub

- Architected a multi-tenant backend with PDF/TXT ingestion, durable PostgreSQL-backed indexing jobs, separate API and worker services, and versioned migrations.
- Combined Qdrant dense-vector search with PostgreSQL full-text search through Reciprocal Rank Fusion and streamed grounded answers with citations over SSE.
- Added JWT authentication, tenant isolation, offline RAG evaluation, Prometheus metrics, request-correlated JSON logs, explicit provider failure mapping, and compensating cleanup.
- Created 118 automated tests covering API contracts, authentication, retrieval, streaming, metrics, and failure paths; CI runs linting, dependency audit, integration tests, and a production Docker build.

EDUCATION AND CREDENTIALS

Bachelor's in Information Systems and Technologies (F6) | Danube Institute of National University 'Odesa Maritime Academy' | Expected 2030

Certifications Database Integration and Management - Microsoft; Linux Commands and Shell Scripting Essentials V2 - IBM

LANGUAGES

Ukrainian - Native | Russian - Native | English - Intermediate (B1)